

IAM-08: Multi-Environment IAM

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Scaling IAM Across Multiple Environments

Enterprise organizations typically have multiple Dynatrace environments (development, staging, production). This notebook covers strategies for managing IAM consistently across environments while maintaining security boundaries.

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Prerequisites

Requirement	Details
Dynatrace Environments	Multiple environments under one account

Requirement	Details
Permissions	Account-level IAM admin
Knowledge	IAM-01 through 07 completed

1. Multi-Environment Architecture

Understanding the relationship between accounts and environments is foundational.

Account and Environment Hierarchy

```
Dynatrace Account (Organization Level)
├─ Environment: Production
│   ├── Entities, Logs, Traces
│   └─ Environment-specific settings
├─ Environment: Staging
│   ├── Entities, Logs, Traces
│   └─ Environment-specific settings
└─ Environment: Development
    ├── Entities, Logs, Traces
    └─ Environment-specific settings
```

Permission Scopes

Scope	What It Controls	Example
Account	Users, groups, SSO, billing	Create users, manage IdP
Environment	Data access, settings, configurations	Read logs, modify alerting
Segment	Subset of environment data	Access only prod-frontend data

Common Environment Patterns

Pattern	Environments	Use Case
Standard	Dev, Stage, Prod	Software development
Regional	US, EU, APAC	Geographic compliance

Pattern	Environments	Use Case
BU-based	Retail, Finance, HR	Business unit isolation
Tenant	Customer A, B, C	MSP/multi-tenant

2. Account-Level vs Environment-Level Permissions

Understanding where permissions are managed is critical for proper access control.

Account-Level Permissions

Managed in Account Management, apply across all environments:

Permission	Description
<code>account-iam-admin</code>	Full IAM administration
<code>account-viewer</code>	View account information
<code>account-company-admin</code>	Manage account settings
<code>account-env-admin</code>	Create/manage environments

Environment-Level Permissions

Scoped to specific environments:

Permission	Description
<code>storage:logs:read</code>	Read logs in environment
<code>storage:metrics:read</code>	Read metrics
<code>settings:objects:read</code>	Read configuration
<code>settings:objects:write</code>	Modify configuration

Permission Assignment Matrix

Role	Account Perms	Prod	Stage	Dev
IAM Admin	account-iam-admin	-	-	-
Prod Viewer	-	Read	-	-
Prod Editor	-	Read/Write	-	-
Dev Team	-	Read	Read/Write	Full
Platform Team	-	Full	Full	Full

3. Centralized vs Federated Models

Choose a governance model based on organization structure.

Centralized Model

Single IAM team manages all access.

Aspect	Approach
Group creation	Central team only
Policy authoring	Central team only
Access requests	Ticketing system
Approvals	Central team

Pros:

- Consistent standards
- Single point of control
- Easier compliance

Cons:

- Bottleneck for requests
- Slower response time
- Less team autonomy

Federated Model

Delegated administration to teams.

Aspect	Approach
Group creation	Team leads (within namespace)
Policy authoring	Central team defines templates
Access requests	Direct to team lead
Approvals	Team lead + central audit

Pros:

- Faster access provisioning
- Team ownership
- Scales with organization

Cons:

- Requires governance framework
- Risk of inconsistency
- More complex auditing

Hybrid Model (Recommended)

Responsibility	Owner
Account-level IAM	Central team
Production access	Central team
Non-prod access	Team leads
Policy templates	Central team
Custom policies	Team leads (with approval)

4. Cross-Environment Access Patterns

Define how users access multiple environments.

Pattern 1: Environment-Specific Groups

Separate groups per environment.

Group	Environment	Permissions
dt-prod-viewers	Production	Read-only
dt-stage-editors	Staging	Read/Write
dt-dev-admins	Development	Full

Use when: Strict environment isolation required.

Pattern 2: Role-Based Groups

Same role, different environment permissions.

Group	Prod	Stage	Dev
dt-sre-team	Editor	Editor	Editor
dt-dev-team	Viewer	Editor	Admin
dt-security-team	Viewer	Viewer	Viewer

Use when: Consistent role across environments with varying access levels.

Pattern 3: Escalation-Based Access

Production access requires explicit escalation.

Group	Normal Access	Escalated Access
dt-oncall	Stage/Dev	Production (during incident)
dt-deploy	Stage	Production (during deploy window)

Use when: Production access is exception, not default.

5. Break-Glass Procedures

Emergency access when normal procedures can't be followed.

What is Break-Glass?

Emergency access mechanism for critical incidents when:

- Normal approvers unavailable
- SSO/IdP is down
- Critical incident requires immediate elevated access

Break-Glass Components

Component	Purpose
Emergency account	Local account (not SSO)
Elevated permissions	Admin or full access
Secure storage	Password vault, split secrets
Audit logging	All actions logged
Post-incident review	Mandatory after use

Break-Glass Setup

1. Create emergency account

- Local Dynatrace account (not SSO-dependent)
- Strong, complex password
- MFA with hardware token

2. Assign minimal necessary permissions

- Typically `account-iam-admin` + environment admin
- Or scoped to specific emergency needs

3. Secure the credentials

- Store in enterprise vault (CyberArk, HashiCorp)
- Or use split-knowledge (2+ people needed)
- Document retrieval procedure

4. Define usage criteria

- When break-glass can be used
- Who can authorize
- Notification requirements

5. Establish post-incident process

- Rotate credentials after each use
- Review all actions taken
- Document in incident report

Break-Glass Policy Template

BREAK-GLASS ACCESS POLICY

Authorized Situations:

- ☐ P1/P2 incident AND normal approvers unavailable
- ☐ SSO/IdP outage preventing authentication
- ☐ Security incident requiring immediate response

NOT Authorized:

- ☐ Convenience (approver busy but reachable)
- ☐ Non-urgent requests
- ☐ Bypassing normal approval process

Usage Steps:

1. Attempt normal access/approval first
2. Document why break-glass is needed
3. Notify security team
4. Retrieve credentials from vault
5. Perform necessary actions only
6. Log out immediately when done
7. Notify IAM team for credential rotation
8. Complete post-incident review within 24h

6. Synchronizing IAM Configuration

Keep IAM configuration consistent across environments.

Configuration as Code

Store IAM configuration in version control:

```

iam-config/
├─ groups/
│   ├── platform-team.yaml
│   ├── dev-team.yaml
│   └── security-team.yaml
├─ policies/
│   ├── environment-viewer.yaml
│   ├── environment-editor.yaml
│   └── log-reader.yaml
└─ boundaries/
    ├── production-boundary.yaml
    └── sensitive-data-boundary.yaml

```

Synchronization Approaches

Approach	Pros	Cons
Manual	Simple	Error-prone, doesn't scale
Terraform	Version controlled, repeatable	Learning curve
Dynatrace API	Direct control	Custom scripting needed
Monaco	Dynatrace-specific	Configuration complexity

Change Management Process

Step	Action	Approval
1	Create PR with IAM change	Developer
2	Review configuration	IAM team
3	Apply to dev environment	Automated
4	Test in dev	Developer
5	Apply to staging	Automated
6	Validate in staging	IAM team
7	Apply to production	Manual approval + deploy

7. Multi-Tenant Considerations

Special considerations for MSPs and multi-tenant deployments.

Multi-Tenant Architecture

Model	Isolation Level	Use Case
Separate Accounts	Highest	Different legal entities
Separate Environments	High	Business unit isolation
Boundaries within Environment	Medium	Team isolation
Segments only	Lower	Data filtering only

Tenant Isolation Checklist

- ☐ Data cannot be accessed across tenants
- ☐ Users cannot see other tenants exist
- ☐ Configuration cannot impact other tenants
- ☐ Audit logs are tenant-specific
- ☐ Billing is tenant-specific

MSP Considerations

Aspect	Recommendation
Customer environments	Separate environment per customer
MSP admin access	Dedicated MSP admin group
Customer admin access	Scoped to their environment only
Audit	Separate audit logs per customer
Compliance	Document isolation in SOC2

Monitoring Multi-Environment IAM

Query patterns for multi-environment oversight.

```
// Cross-environment access attempts
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "environment")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Account-level administrative actions
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "account") and (matchesPhrase(content,
"admin") or matchesPhrase(content, "created") or matchesPhrase(content,
"modified"))
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Break-glass or emergency account usage
fetch logs, from: now() - 90d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "emergency") or matchesPhrase(content,
"break-glass") or matchesPhrase(content, "breakglass")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

Next Steps

With multi-environment IAM understood, proceed to troubleshooting:

Recommended Path

1. **IAM-09: Troubleshooting Access Issues** - Debug permission problems systematically

Multi-Environment Checklist

Before moving on, ensure you have:

- [] Documented all environments and their purposes
 - [] Defined governance model (centralized, federated, hybrid)
 - [] Created environment-specific access groups
 - [] Established break-glass procedure
 - [] Implemented IAM configuration as code
 - [] Documented multi-tenant isolation (if applicable)
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Summary

In this notebook, you learned:

- Multi-environment architecture and hierarchy
 - Account-level vs environment-level permissions
 - Centralized, federated, and hybrid governance models
 - Cross-environment access patterns
 - Break-glass procedures for emergency access
 - IAM configuration synchronization approaches
 - Multi-tenant considerations for MSPs
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References

- [Account Management](#)
 - [Environment Management](#)
 - [Dynatrace Terraform Provider](#)
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